

American University of Armenia

College of Science and Engineering

Capstone

Automated Seasonal Color

Classification and Makeup Recommendation

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Jemma Asryan: Skin detection

Mariana Sargsyan: Other facial features detection

Sona Khachatryan: Seasonal Analysis

Abstract

We propose an automated data-driven seasonal color profile classifier that generates personalized palette suggestions by extracting color-based features from face images. [1]. The project aims at the implementation of a virtual makeup try-on tool that will allow users to apply the recommendations in real time. This research offers a scalable solution for personalized color analysis, bringing image processing, computer vision, color theory, fashion, and cosmetic science together to support individuals in making more confident and color-informed choices in personal style and makeup. It is addressed in four directions:

- Detection of skin-related colored pixels in different color spaces using already-established color theory and image processing methods to enable the segmentation of different skin regions across varying lighting conditions [2];
- Formation of a skin-specific color palette using the extracted skin regions and identification of non-skin facial features such as the eyes, lips, and hair [3]. These facial components are separately color-clustered to form a unique palette representing that individual's natural coloring. Based on the observed color patterns, such as the skin undertone (cool, warm, neutral), the contrast level (high, medium, or low) of facial features, and the clarity (clear or soft), the person's seasonal type (e.g., Soft Summer, True Autumn, Deep Winter, Clear Spring) is identified;
- Finally, the seasonal types are mapped to makeup products to generate personalized recommendations and best-suited clothing color solutions [4].

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1. Introduction

In our increasingly digital world, technology has become an integrated part of many aspects of our daily life, including personal style and beauty. As a result, understanding the automation process of personalized recommendations for makeup and fashion, based on an individual's distinct facial features and colors, has become a significant area of study. This research explores the process of creating a data-driven approach for generating personal color palettes and makeup recommendations based on color theory and the interpretation of color-based facial features. One of the ways of finding the most suitable color palettes and makeup product colors is by finding out one's seasonal type with the help of personal color analysis. This technique finds an individual's seasonal type based on their natural skin tone, eye and lip color, and hair shade. Traditionally, this classification has been based on human perception and, in most cases, is performed manually by fashion or cosmetology experts. However, these rough estimations are subjective and inconsistent. Thus, the automated solution will help to get more accurate personalized color profiling. This capstone project aims to enhance the user's understanding of how specific colors interact with their unique skin and facial features.

One of the key benefits that this research gives to future users is their confidence in making personalized decisions. By knowing their seasonal color type and best-suited color palettes for clothing and makeup, individuals will be able to make style decisions that genuinely complement and enhance their natural appearance. As a result, individuals will be able to enhance their looks in a more harmonious way. Another significant advantage is the time and cost efficiency. By knowing one's seasonal color profile, and being able to virtually try on the

products before purchasing them, users can avoid purchasing products that ultimately go unused because of not suiting their natural color profile. The personalized color guidance will lead to smarter purchase choices, which will avoid wasting products and important resources such as time and finances. This project also promotes accessibility of expert-level color analysis to a wider audience. Traditionally, these services have been offered by stylists or fashion consultants, often at a high cost. The automated tool will enable users to do their personal color analysis by just using their smartphones without the need for high-paid experts, regardless of their financial budget.

This research greatly contributes to the field of image processing by developing methods to extract different facial objects with the help of colors under various lighting conditions. Additionally, the research project offers an innovative way of bringing the traditionally subjective and manual methods of conducting color analysis closer to technology by offering a more accurate, data-driven approach and a systematic and reproducible alternative. By relying on measurable features extracted from facial images, the system reduces human bias and inconsistency, which will consequently increase the reliability of color analysis results.

Another very vital impact of the research is that it brings computer science closer to fields such as color theory, fashion, cosmetics, art, filmmaking, and human perception studies. This interdisciplinary application can enhance our understanding and use of visual principles in real-world contexts, helping to drive innovation not only in technology but also in gaining new artistic insights, and eventually expanding the possibilities for innovation across various industries.

2. Literature Review

Color Models and Their Relevance in Image Processing

RGB color space is one of the most common color spaces for computer imaging and display devices, such as cathode ray tube (CRT) monitors, is the RGB (Red, Green, Blue) color space. It defines colors as different intensities of red, green, and blue light. The triplet value set of these three base colors describes every pixel of an image. Though RGB is simple to implement in hardware and straightforward to carry out digital image processing, its channels are perceptually non-uniform and strongly correlated. This could be troublesome for color-based classification and analysis, especially in applications where meaningful interpretation of color information is intended.

Hue is the prevailing wavelength of color and a key element of color models like HSV (Hue, Saturation, Value) and HSL (Hue, Saturation, Lightness). Hue determines the kind of color seen, for example, red, green, or blue, and is normally expressed as an angular location on a color wheel (0° to 360°). Hue is particularly significant in color discrimination tasks since it records color type regardless of light and levels of saturation.

The HSB (Hue, Saturation, Brightness) color system is meant to be more understandable to human eyesight than RGB. It is composed of three components. Hue is a type of color and is quantified in terms of degrees on a circular color wheel. Saturation refers to how strong or how pure the color is, with low saturation tending towards grayscale and high saturation being a vivid color. Brightness (or Value) is the lightness or darkness of the color. Due to its perceptual

importance, the HSB model is commonly used in image processing applications such as skin color detection, object segmentation, and visual-based classification. The separation between chromatic and intensity information allows for color analysis that is both computationally more efficient and human interpretable.

Luminance, which is also called brightness or lightness, quantifies the intensity of perceived light from an image. In certain models such as YCrCb and Lab, luminance is decoupled from chrominance (color) signals in an attempt to more closely mimic human visual perception. It is often computed as the weighted sum of the RGB channels with the green channel receiving greater preference owing to the sensitivity of the eye to green light.

$$L = (0.2126 \times R_1) + (0.7152 \times G_1) + (0.0722 \times B_1)$$

Fig. a: Luminance

$$\text{Contrast Ratio} = (L_{\text{light}} + 0.05) / (L_{\text{dark}} + 0.05)$$

Fig. b: Luminance

Seasonal Analysis: Introduction & Historical Background

Later, in the 1950s, Suzanne Caygill refined this idea by developing personalized palettes for clients based on personality, appearance, and energy, and created the first formal version of

the 4-season system. In the 1980s (Caygill, 2022), Carole Jackson popularized it for the public through her book *Color Me Beautiful*. She taught that by knowing your season, you can look more vibrant and confident with less effort (Jackson, 1987).

As the system evolved, analysts like Kathryn Kalisz expanded it into 12 subtypes — like Soft Summer, Deep Autumn, or Bright Spring — using color science and standardized tools (Chrysalis Colour, 2019).

- Then talk about the 4 features needed to calculate them
- Then explain how each feature is scaled from 1-5
- Explain how we get the needed values and normalize them in a scale of 1-5 from all the face palettes
- Then go over each of the 16 Seasonal types and give their characteristics,
- Then show the table that determines the seasonal type
- Talk about the product recommendation
- How it works

(Lit. review) Seasonal Color Analysis: Definition & Historical Background

Seasonal color analysis is a categorizing technique used to determine a person's ideal color palette based on their natural features, such as skin tone, hair color, eye color, and lip color. The idea of dividing colors into seasonal palettes first originated in 1961, by Johannes Itten in his book *The Art of Color* (Itten, 1961). As an artist and a teacher, Itten noticed how his students painted better when using colors of their choice. Based on this, he suggested that using personal colors leads to a more vibrant work; therefore, he states that knowing one's personal color palette

is crucially important. He began dividing persons into warm and cool groups based on natural color and correlating that to seasonal color palettes: Spring, Summer, Autumn, and Winter , describing each season's colors the following way:

(Itten, 1961, pp. 24-31).

In his book, Itten introduces the concept of dividing colors into 4 seasons: Spring, Summer, Autumn, and Winter, and that an individual's season was based on their nature, personality, and looks (Itten, 1961, p. 128).

The color space Itten considered in the book is a spheric color space consisting of 12 meridians and parallels, representing the Hue, Saturation, and Luminance (HSL) (Itten, 1961, p. 111), (**Fig.1**). Similarly, in our research, the color spaces we considered for seasonal type classifications is the Munsell System, consisting of the Hue, Value/Luminance, and Chroma ((L Caivano, 2015, pp. 4-6) (**Fig.2**).

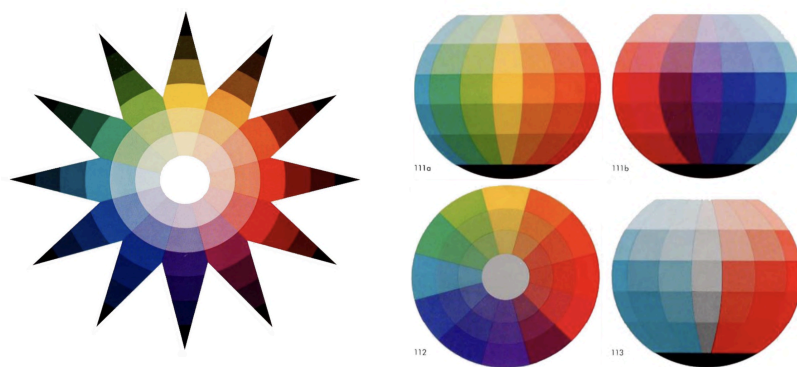


Fig.1: Color Sphere (HSL)

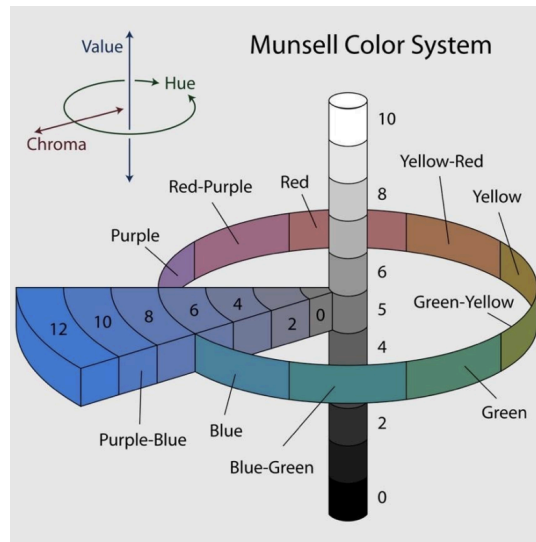


Fig.2: Munsell Color System (*L Caivano, 2015*)

Later, in the 1950s, Suzanne Caygill furthered the idea with palettes for specific clients based upon personality, appearance, and energy and codified the basic 4-season method. Subsequently, in the 1980s, (Caygill, 2022) it became mainstreamed by Carole Jackson in her book *Color Me Beautiful*. She instructed that you could simply look more alive and confident by knowing what your season is (Jackson, 1987).

Later, scientists such as Kathryn Kalisz continued to refine it into 12 subtypes – such as Soft Summer, Deep Autumn, or Bright Spring – based on color science and quantitative measures (Chrysalis Colour, 2019).

3. System Description

Traditional seasonal color analysis, a method for understanding the most suitable colors for a person based on their facial features, is generally subjective, non-reproducible, and is mainly based on subjective human perception. The process is usually conducted manually, which makes it prone to bias. In the present era of digitization and information, the possibility of making personalized beauty advice more accurate and data-driven becomes more realistic.

Although image processing and computer vision have advanced, there are still not many systems that objectively evaluate facial color attributes like undertone, contrast, intensity, and value for seasonal color classification. Moreover, the majority of digital applications do not provide built-in product recommendations.

This study addresses those problems by creating automated methods and suggesting algorithms for accurate face feature color detection and seasonal analysis. The goal is to take a person's facial image as input, extract the skin, lip, eye, and hair color, and conduct seasonal analysis, which will classify them into one of the 16 seasonal types. Based on that classification, the system should recommend the most suitable color palette for that person, and also give makeup product recommendations. This approach will help to reduce human subjectivity for personalized color analysis.

Technology

Our aim for this project is to use image processing tools to analyze face images, detect facial features, and extract their colors. To do that, we use the ImageJ program. ImageJ is a Java-based image processing program that can solve many image processing and analysis problems via user-written Java plugins and recordable macros. ImageJ is used in this project to segment and separate facial features like skin, eyes, lips, and hair, and conduct analysis on the features identified.

Dataset

The dataset used to conduct the research consists of 200 face images taken under varying lighting conditions to test the developed algorithms and image processing techniques(Carlos Eduardo Thomaz - Personal Web Page, 2010). In the future, the following dataset can be used to train supervised machine learning models as well. This dataset allows us to test our experiments before choosing the best method for the final product.

Makeup products database

We use this database to recommend the products(Cosmetic Brand Products Dataset, 2023).

4.a Methodology and Solutions:

Chapter 1: Skin Detection

Our research starts with the detection and isolation of a person's facial features, such as skin, eyes, lips, and hair, via image processing tools(reference to Background section).

The first and perhaps the most critical part is to accurately detect skin regions from the person's face image, which directly affects the precision of the color analysis. Therefore, we conduct pixel-wise analysis using the HSV(Hue, Saturation, Value) color space to isolate skin regions. To determine skin pixels, we evaluate each pixel's HSV values and check if each their hue and the product of saturation and brightness fall within the predefined range.

Skin detection criteria are based on two main conditions:

1. Hue value should be within a certain range typical of human skin tones.
2. The product of saturation and brightness/value, scaled by 256, should fall within a specified range. ($s * v * 256$)

To evaluate the effectiveness of different ranges, we conducted several experiments. We started by testing broad ranges for Hue, such as 0-30, based on general references from skin detection literature(Journal, 2013). In this test, the hue range includes more skin (from the shoulders). For the Saturation * Brightness value, we also experimented with various thresholds, starting from a wide range, such as 30-150. It includes lighter shadows, more pixels from brows and hair. The combination of hue and the product of saturation and brightness resulted in more skin-like pixels, more shadow pixels, and more included lips pixels.

For understanding the best range for the $S * V$ value, we use several experimental ranges. One of the most promising ranges was (45 - 105). This range was effective in isolating undesired hair regions, especially the beard for men. However, this range had the disadvantage of removing too many skin pixels around the eyes, which actually play a great role in identifying one's undertone. Another range tested was (35 - 105). While also promising, and helping to exclude the hair, this range kept some parts of the lips, which was not the original goal.

Finally, we started experimenting with the combinations of the hue and sv ranges. One of the most interesting cases was the range of hue (3 - 21) and sv (35 - 80). Even though it was very close to the range that we chose in the end as the most optimal one, this combination of ranges was an extreme case. This combination filtered out everything except lips, and a very small part of the skin around the nose. Another range tested was hue (5 - 30) and sv (29 - 170). This range kept some parts of the lips, which were not desired. Additionally, for the skin pixels, only the pixels at the central part of the face were kept, which could again lead to inaccuracies when trying to find the desired colors.

We continued experimenting and tried different ranges both for hue, SV, saturation and brightness separately, and all possible combinations, in different lighting environments.

During our research, we also had a goal of developing a method of finding the most optimal hue and saturation * and brightness range for each person. Some of the results of the previous method tested showed that there are certain ranges which work well for some part of the dataset but not on the other, and vice versa. We tried to create a method that will be personalized for every person. The method in question was to set an initial range, filter the skin pixels with

that range, calculate their average, and then change the range according to the average value. Then the plugin should have been run on the original image one more time, and the same process should be repeated until the average value of pixels converges to a certain value. There were multiple methods tested to put this theory into action.

Method 1: Recursive

For the first method for creating personalized filtering ranges, we implemented a custom Java plugin. The plugin had a similar logic to our original plugin used to filter the skin in the HSV color space. The code operates by filtering pixels that fall within a specific range of hue and combined saturation-value (SV) levels. The plugin begins by duplicating the input image and setting an initial hue range of 0 to 30 degrees, which covers common skin tones in the hue spectrum. The SV range is initially fixed between 40 and 80, representing moderate saturation and brightness levels suitable for skin. The pixels satisfying the mentioned hue and SV ranges are filtered, and all other pixels are set to white. The core feature of this plugin is its recursive function, which helps to recursively change the hue range based on the previous one. After filtering each image, the average hue of the remaining pixels is calculated. This average is then used to refine the hue range for the next iteration by narrowing it around the mean hue (± 15). We chose 15 to keep the same size as the initial range, which was 30. The recursion continues until the hue range stops changing the value of the previous iteration.

The same method was tried with keeping the hue range unchanged and changing the sv range recursively. Afterwards, we made both ranges change recursively simultaneously, and stopped the recursion when both ranges stopped changing their values. Because of the range being

updated by keeping the same distance, the accuracy of our results decreased. The plugin started including some parts of hair, eyebrows, and lips, and excluded too much area from the skin region around the eyes. So we decided to implement another method of changing the ranges recursively by using the standard deviation instead of the average.

Method 2: Recursive

We enhanced the initial version of the skin detection filter with the help of dynamic recursive ranges, and we introduced a new plugin that included a more statistically accurate method to refine the hue range. This filter changes the range based on statistical variance and standard deviation, resulting in a more accurate and robust skin segmentation process. The algorithm begins with an initial hue range (0–30) and sv range (40-80), which approximates typical skin tones. The mean hue (μ) and standard deviation (σ) of retained pixels are computed. A dynamic adjustment factor k is applied, where:

1. $k=1$ if $\sigma < 5$
2. $k=1.5$ if $\sigma < 10$
3. $k=2$ otherwise.

The new hue range becomes $\text{HueMin} = \mu - k\sigma$, $\text{HueMax} = \mu + k\sigma$

This approach ensures the filter adapts to the actual distribution of hues in each image, improving the accuracy. In this method, the recursion terminates when the range becomes too small (less than 7 units wide), ensuring the algorithm does not overfit or exclude valid skin areas. In most cases, this method was better at excluding non-skin features, but at the same time, it excluded skin pixels that were not supposed to be excluded, specifically on the cheeks and the

outer border of the face. As we did for the previous method, here we also tried keeping the hue range unchanged and changing the sv range recursively, and then updating both ranges simultaneously.

Both of the methods described above were also tested on images having darker lighting conditions than usual. The same process was applied with a different initial range of hue (0-30) and sv (40-96).

After conducting all of the experiments mentioned above, we came to the conclusion that in a standard lighting environment, the best ranges for filtering the skin pixels are the following:

1. Hue: 3 - 24
2. S * V: 40 - 80

The same experiments were conducted on the facial images with darker lighting conditions. The most effective ranges found were:

1. Hue: 3 - 24
2. S * V: 30 - 96 (can be approximated to 100)

Palette Creation

The next stage is creating color palettes from extracted skin pixels from face images. Extracted skin pixels are divided into groups by hue, saturation and brightness/value. Again, we

did various experimental groupings to see which one gives more information about the actual skin color.

Method 1:

The first approach was to create a skin color palette in two steps:

1. Divide extracted skin pixels into three equal main color groups based on their hue. The reason behind this approach was to separate different underlying colors of skin, such as reddish, yellowish, or greenish. The first group included reddish tones, the last group included yellowish/greenish tones, and everything else was included in the middle group.
2. In each of these three main hue groups, divide pixels into two equal subgroups by the product of saturation and brightness. The reason for doing this is to separate very bright or vibrant skin tones from darker or less saturated hues.
3. To get the final skin color, we average this 6-color palette.

Method 2:

The second method is again a two-step process:

1. Now, divide the extracted skin pixels into two equal main hue groups: one group included 3 to 12 hues, the second group included 12 to 24 hues. This way, we strictly put reddish colors in one group, and yellowish/greenish colors in another group.
2. Within each of these two equal hue groups, we divide the pixels into three equal subgroups based on the product of saturation and brightness: high, medium and low brightness. This way, we capture a broader spectrum of skin tones.

3. To get the final skin color, we average this 6-color palette.

Method 3:

Our third method for skin palette creation is slightly different than the first two. First, we scaled down the entire face image, which resulted in fewer skin tone variations because when scaling, pixels are averaged by their surrounding pixels. Then, we extract skin pixels and again create a 6-color palette as we did in the previous methods. In this approach, we again use 3 hue and 2 saturation times brightness and 2 hue and 3 saturation times brightness groupings to compare the results. This approach aimed to get a general overview of the dominant skin tones; however, this caused the loss of some undertone skin pixels, which can change the resulting skin color.

Method 4:

Our final and most effective method for generating a skin color palette involved a more refined approach, aiming to get the most accurate skin tone representation despite challenging image conditions.

1. Divide the skin pixels into five equal groups based on their hue, within the hue range of 3 to 24. This grouping allowed us to capture a good spread of skin tones, from reddish to yellowish.
2. Then, to calculate the final skin color, we take the average of only the last three groups of hue, that is, the range of 13 to 24. The reason behind this is important because we found out that in the typical skin tone, the influence of red pixels (reddish tones), which usually fall into low hue ranges, sometimes disproportionately shifts the average color towards red. So, by the exclusion of these red hue groups, we give less weight to red pixels and

more to yellow pixels. This method resulted in a more accurate and skin-like average color.

3. In addition, because of our dataset's lighting conditions, we saw that the average skin color was a bit darker than the actual skin of the person. Therefore, we increased the brightness of each of these colors from our generated palette by 20%. By changing the brightness of these skin-like hues and then calculating their average, we managed to derive a final skin color that more closely matched the true skin tone, even when faced with challenging lighting variations.

4.b Methodology and Solutions:

Chapter 2: Facial Feature Detection: Eyes, Lips, Hair

After the extraction of the skin region, there is a need to generate a technique that can accurately extract color palettes from key facial regions such as the eyes, lips, and hair. In our research project, we created an automated ImageJ-based macro workflow to segment facial features and extract their dominant colors using HSV filtering, morphological operations, and ImageJ plugins created with the programming language Java.

Eyes and lips extraction

The eye and lip color palettes are one of the most important parts for conducting seasonal color analysis, and especially for product recommendations. To segment the eyes and lips, we perform image processing operations using ImageJ. Below are the respective steps:

1. A duplicate of the original image was created to preserve the original data.
2. The original skin filtering HSV-based color filter was applied to filter out the skin region.
3. The filtered image is converted to 8-bit grayscale to enable thresholding. In order to get a pure black and white image, we applied a threshold of 254, which means that all values greater than or equal to 255 are presented as white and the rest as black.
4. A binary mask is created by automatically thresholding the image, effectively isolating mid-to-dark tonal regions that typically correspond to facial features like lips and eyes.
5. This binary mask is effectively isolating the facial features like lips and eyes, where backgrounds are set to black, while feature regions are set to white.

6. “Fill holes” was applied, which detects the white objects of the binary image. Then, it finds the black pixels that are entirely surrounded by white pixels and identifies them as “holes”. These holes within the binary regions are filled to make the objects solid.
7. The mask is then inverted to prepare the binary image for subtraction, separating feature regions from surrounding noise.
8. With the help of an image calculator, an image subtraction operation is performed to remove filled-in areas from the original binary mask, producing a cleaner segmentation of target facial features(e.g., eyes, lips).
9. The refined mask is dilated to get rid of undesired small pixel regions, which do not represent any of our targeted facial features.
10. The resulting mask is subtracted from the original image to suppress the background and isolate the real pixels of the original image of our desired areas (e.g., eyes and lips).

After effectively extracting eyes and lips, we aim to generate color palettes to select the most accurate eye and lip colors. To create these palettes, our research progressed by exploring methods to detect eye and lip regions correctly.

Eyes and lips region detection: Method 1

The first approach relied on identifying non-skin pixels and using their spatial distribution to estimate feature locations. By calculating the average y-coordinate of the non-skin pixels on unique y coordinates, a horizontal axis was established to separate the upper (eye) and lower (lip) regions. Pixels located above this y-axis were assumed to belong to the eyes, and pixels below the y-axis belonged to the lips.

From the upper region, we get saturation and brightness (value) from each pixel. To analyze saturation, we calculate the minimum, maximum, and middle saturation values. Then, pixels are categorized into low saturation and high saturation groups. Next, we categorize upper region pixels by their brightness(value). We divide them into low, middle, and high brightness groups. Finally, to represent the final set of eye colors, we combine all pixels from the low saturation group and both the low and mid brightness groups. As a result, we generate an eye color palette with 4 colors. The first and the last colors of this palette are usually black and white, if this is the case, we calculate the average eye color by the remaining two colors.

To isolate lip-colored pixels, an RGB-based filter was applied: pixels satisfying the condition $(R + B) / 2 > G$ were selected, targeting the characteristic reddish pigmentation of lips by emphasizing magenta over green. Multiple aggregation strategies were tested on the selected pixels, including simple average, weighted average, and median, to generate a representative lip color. However, the resulting colors consistently skewed darker and more brown than the lips' actual appearance. This discrepancy was largely due to the inclusion of shadowed areas and the natural variability in pigmentation across different parts of the lips. Our analysis revealed that lip color varies significantly within the lip area, with inner, middle, and outer sections displaying distinct brightness levels. To capture this gradient, the filtered lip pixels were grouped into three clusters based on brightness, and an average color was computed for each cluster. Here is the process:

1. A brightness threshold was introduced: only pixels with brightness greater than 50 were retained.
2. Using the luminance formula $0.299 * R + 0.587 * G + 0.114 * B$, the brightness of each pixel was found. And the filter $Brightness > 50$ was applied to exclude dark shadows on

the lips. The value 50 was chosen because it removes only the darkest 20% of possible brightness values, which avoids losing midtones or highlights, which are critical for creating an accurate lip palette.

3. The resulting filtered lip pixels were sorted by brightness.
4. The sorted pixels were divided into three equally sized groups of low brightness, medium brightness, and high brightness.
5. For each of the three groups, the average color is computed using HSB space (Hue averaging is handled using vector math to avoid circular hue artifacts).
6. Each average color is rendered as a new rectangle in a 3-color palette.

This filtering helped exclude darker, shadowed pixels and improved color accuracy at some scale.

Finally, to improve interpretability and separate chromatic information from luminance, all RGB values were converted to HSV (Hue, Saturation, Value) space. This allowed for a more meaningful representation of the extracted colors in the context of personal palette generation.

On the other hand, this method of eye and lip detection suffered from false positives since some non-eye pixels above the y-axis and some non-lip pixels from below the y-axis were included as the filtered pixels in the grouped palettes. Therefore, we proposed another method of accurate eye and lip region detection.

Eyes and lips region detection: Method 2

To overcome this limitation, the methodology shifted from identifying pixels based on the average y-coordinate to region-based segmentation. After excluding the skin region, the

remaining pixels were clustered to identify contiguous regions. Each region was evaluated based on its spatial coordinates (x and y) and area. The regions with the two largest areas and lowest vertical centroid (i.e., closest to the top of the face) were classified as the eye region. The region with the highest vertical centroid (i.e., closest to the bottom of the face) was classified as the lip region, assuming a standard upright facial orientation. The refined color extraction process was then applied exclusively to the identified eye and lip regions.

Hair region detection

The second method described was used for detecting the eyes and lips serve as a base for identifying the hair region and the dominant hair color.

The hair region detection process begins with the detection of the eye regions in the face described above. Using the bounding box of the eyes, the image is cropped from the upper-left Y-coordinate + 20, effectively capturing the area just above the eyebrows. This ensures that the hairline region is included while avoiding unnecessary facial features. After the image is cropped we create two copies and apply different filters on each:

1. **Skin Region Extraction:** On the first copy, a color filter is applied using the original skin filtering plugin with a hue range of 3–24 and an SV (saturation × value) range of 40–80. This combination, as already described, effectively filters the skin region of the image.
2. **Skin + Hair Region Extraction:** On the second copy, only the hue filter (3–24) is applied, which retains both the skin and hair regions. This is also a great demonstration that the filtering of hair is highly dependent on the SV value.

3. Both filtered images are then converted to binary format and processed with the "Fill Holes" operation in ImageJ. This step ensures that the maximum possible hair region is included in the mask.
4. To isolate the hair region, the binary image from the first copy (skin only) is subtracted from the binary image of the second copy (skin + hair). The result is a binary mask that represents the hair region alone.
5. Finally, this binary hair mask is used to subtract from the original cropped image, producing an image containing only the hair region. This output can then be used for further color analysis or palette extraction tasks.

After isolating the hair region, a new ImageJ plugin is used to extract a single representative color for the hair. The plugin processes the hair image by examining each pixel and selecting only those that are neither pure black nor pure white, as these colors represent the binary masked part. This way, we consider only the pixels of the identified hair region. The plugin then converts these selected pixels into the HSB color space. This conversion allows for a more perceptually accurate calculation because hue is circular in nature, which means it wraps around at both ends of its scale, and is therefore averaged using trigonometric functions by summing the cosine and sine components of the hue angles and calculating their average direction. Saturation and brightness values are averaged linearly. After computing the average hue, saturation, and brightness, the plugin converts these values back to RGB to obtain the final average hair color. This approach finds the most dominant hair color of a person from their facial image and outputs the color in a tile.

4.c Methodology and Solutions:

Chapter 3: Seasonal Analysis and Product Recommendation

Methodology & Image Processing

In our research, we concluded that seasonal color analysis is based on the Munsell Color System ([See Fig.2](#)), using 4 main factors:

1. *Hue/Undertone (cool or warm): (spectre: 0° - 360°),*
 2. *Intensity/Chroma (soft or vivid): (spectre: 0 - 1),*
 3. *Value/Brightness (dark or light): (spectre: 0 - 1),*
 4. *Contrast ratio (difference of luminance (L) between facial features): (1:1 - 1:21). ([See Fig.a,b](#))*
-

Each of the facial features we extracted: skin, hair, eyes, and lips, that we got in the previous steps from the generated palettes, are used in the calculation of all 4 factors in the following way:

-
1. **Hue** = Hue of skin color;
 2. **Chroma** = (Chroma of Skin + Chroma of Eyes + Chroma of Lips) / 3,
 3. **Value** = (0.7 * Value of Skin) + (0.15 * Value of Eye) + (0.15 * Value of Hair),
 4. **Contrast Ratio** = (Lmin + 0.05) / (Lmax + 0.05)

[Lmin and Lmax are picked from the minimum and maximum values of the Luminances of skin, hair, and eyes, calculated with the formula [Fig.a,b](#)]

In order to obtain a uniform calculating scale, we normalized each of these 4 factors into a scale from 1-5, dividing their ranges into 5 respective parts.

1. Hue (Scaled 1-5)

Here, in the code below, we can observe the different ranges defined for the skin hues, scaling from coldest to warmest. The ranges were picked by setting the colors with more Yellow to be the warmest, and the ones with more Red and Pink to be the coldest, and the range is later divided into 5 equal parts. These assumptions are based on the research, discussing how to determine the temperature of the colors (Itten, 1961, p. 61), (Sirisayan, n.d., p. 16).

```
private int normalizeHue(float hue) {  
  
    if (hue < 8) return 1;      //coldest (4-8)  
  
    if (hue < 12) return 2;    //cold (8-12)
```

```
    if (hue < 21) return 3;    //neutral (12-21)

    if (hue < 27) return 4;    //warm (21-27)

    return 5;                  //warmest (27-33)
}
```

2. Chroma (Scaled 1-5)

```
private int normalizeChroma(float chroma) {

    if (chroma < 0.2f) return 1;    //muted (0-0.2)

    if (chroma < 0.4f) return 2;    //soft (0.2-0.4)

    if (chroma < 0.6f) return 3;    //moderate (0.4-0.6)

    if (chroma < 0.8f) return 4;    //saturated (0.6-0.8)

    return 5;                      //brightest (0.8-1.0)
}
```

3. Value (Scaled 1-5)

```
private int normalizeValue(float value) {

    if (value < 0.2f) return 1;    //lightest (0-0.2)

    if (value < 0.4f) return 2;    //light (0.2-0.4)

    if (value < 0.6f) return 3;    //medium (0.4-0.6)

    if (value < 0.8f) return 4;    //dark (0.6-0.8)

    return 5;                      //darkest (0.8-1.0)
}
```

4. Contrast (Scaled 1-5)

```
private int normalizeContrast(float contrast) {  
    if (contrast < 5f) return 1;    //lowest (0-5)  
    if (contrast < 9f) return 2;    //low (5-9)  
    if (contrast < 13f) return 3;   //medium (9-13)  
    if (contrast < 17f) return 4;   //high (13-17)  
    return 5;                       //highest (17-21)  
}
```

Now, based on this classification, we have come up with a system to determine a person's seasonal type out of the 12 main and 4 transitional seasonal color types, basing our conclusions on the literature referring to the most common characteristics of each of the seasonal types (Fig.3). For instance in *Color for Men, Determining your Season*, it is mentioned that people with Winter seasonal palette have skin with cool undertones and lower intensity (having more grey undertones), their features have an overall higher contrast ratio (light skin and dark hair / light eyes and dark hair) (Jackson, 1987, pp. 50-54). That is why for the 3 main subtypes of Winter: Bright Winter, True Winter, Deep Winter, we can notice the high values of Contrast, moderately high Value, low chroma, and cool undertones (See Fig.3).

Feature/ Season	Soft Summer	True Summer	Light Summer	True Light	Bright Spring	Light Spring	Warm Spring	True Bright
Undertone	2	1	2	3	4	4	5	3
Intensity	5	5	4	3	1	2	2	1
Value	3	3	5	5	2	5	4	3
Contrast	2	2	3	3	5	4	4	5

Feature/ Season	Bright Winter	True Winter	Deep Winter	True Deep	True Autumn	Deep Autumn	Soft Autumn	True Soft
Undertone	2	1	1	1	3	4	4	3
Intensity	1	1	1	2	2	4	5	5
Value	4	2	2	1	1	2	3	3
Contrast	5	4	3	3	3	2	2	2

Fig.3: Classification of 16 Seasonal Color Types (based on 1-5 scaling system)

The next step is to determine respective color palettes for each of the seasonal color types. Since each of the seasonal color types has respective characteristics not only for the people's personal features, but also for the complementary colors for each individual. In the book *The Art of Color*, for instance, the Spring color palette is described to be energetic, light, radiant, like yellows and pinks, symbolizing youth and vitality. Or in the book *Color for Men, Determining your Season*, Summer Color palette is said to have colors like: Blue-Gray, Gray Sky, Blue, Rose-Brown, Burgundy, Pink, Soft White (Jackson, 1987, p. 44).

Based on the research we have conducted, we are suggesting a method to determine and display the seasonal color palettes for each season in the following way. We will need to use the 3 factors of Hue, Chroma, and Value, however, opposite to the method described above, determining an individual's seasonal type that only included hues of the range 3-24 (in the scale

of 0-255) or 4-33 (in the scale of 0-360), which are described to be the hues typical to skin colors, as mentioned in Chapter 1, here we will need to use the whole spectre of colors.

To understand what the ranges should be for each seasonal type, we will need to determine the coldest to the warmest shades of the hues. Based on the research, we have concluded that the colors that have more blue in them are cooler, and the ones that have more yellow in them are the warmest (Sirisayan, n.d., p. 6). For that, we have divided the color spectrum into the following 5 equal parts (**Fig.4**).

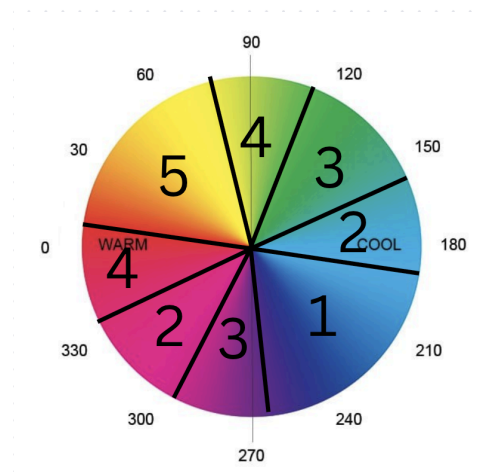


Fig 4: Color Wheel Divisions of Cool and Warm Ranges

Hue Score 5: (9° - 81°] //warmest

Hue Score 4: (0° - 9°] or (81° - 117°] //warmer

Hue Score 3: (117° - 153°] or (297° - 333°] //neutral

Hue Score 2: (153° - 189°] or (261° - 297°] //colder

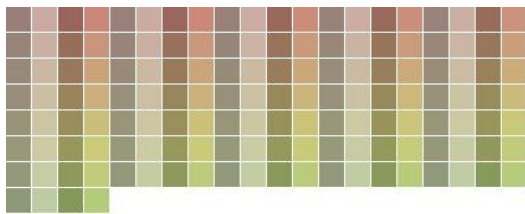
Hue Score 1: (189° - 261°] //coldest

```
private float[] getHueRangeFromScore(int hueScore) {  
    if (hueScore == 5) return new float[]{9f, 81f};  
    if (hueScore == 4) return new float[]{0f, 9f, 81f, 117f};  
    if (hueScore == 3) return new float[]{117f, 153f, 297f, 333f};  
    if (hueScore == 2) return new float[]{153f, 189f, 261f, 297f};  
    return new float[]{0f, 360f}; // For score 1  
}
```

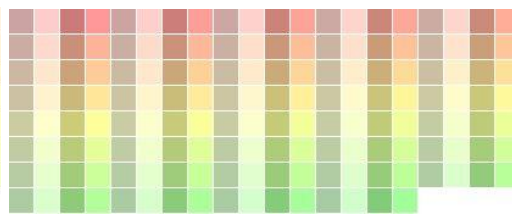
Next, we need to have scores for Chroma and Value as well. But there will be no need for further calculations, as they will have the same 5 equal divisions of ranges, just like in the normalizing methods for seasonal type categorization.

Next, using the same table (Fig. 3) as a reverse map for determining the three scores for each of the seasonal color type, for instance, if a person is determined to be a seasonal color type of *True Autumn*, then we will have the following scores: *Hue* = 3, *Chroma* = 2, *Value* = 1. Now based on these values, it will map to the respective ranges for each of the factors: Hue = (117° - 153°] or (297° - 333°], Chroma = [0.2, 0.4), Value = [0, 0.2). Based on these 3 ranges, all the possible combinations of colors will be kept in an ArrayList (72 x 20 x 20 = 28,800 possible results), and will be used as the color palette for the respective seasonal type. However, to have a more visually appealing presentation of the seasonal color palettes, we have generated the color palettes by displaying the palettes with reduced step sizes of 1/30 for hue values, and 1/20 for chroma and value. Here are all the 16 seasonal color palettes (Fig. 5 - Fig. 20).

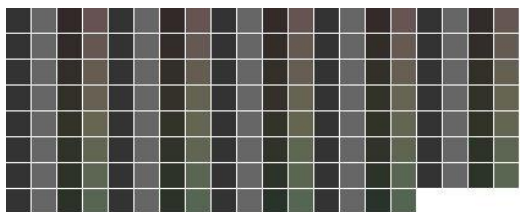
Warm Spring



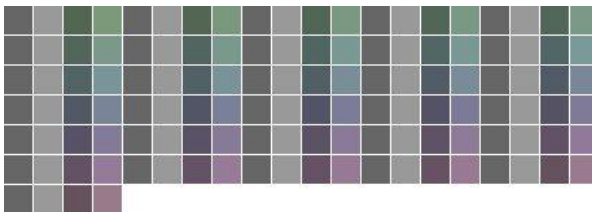
Light Spring



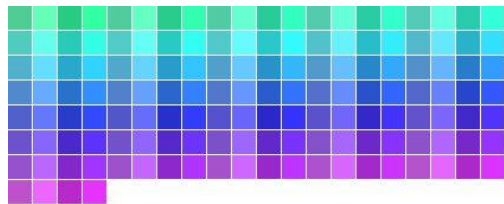
Bright Spring



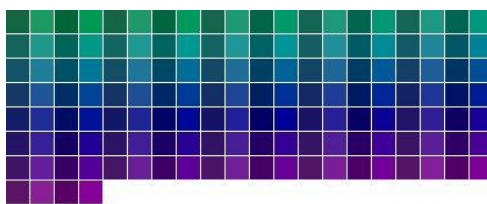
True Bright



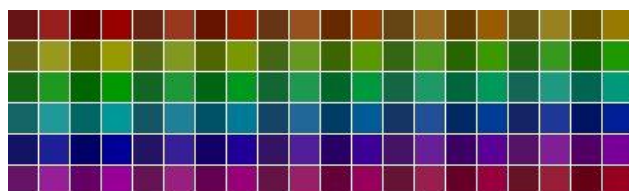
Light Summer



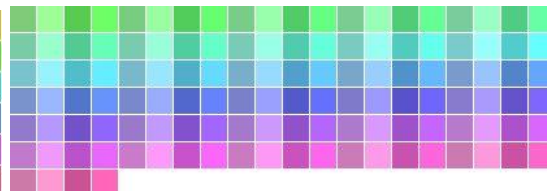
Soft Summer



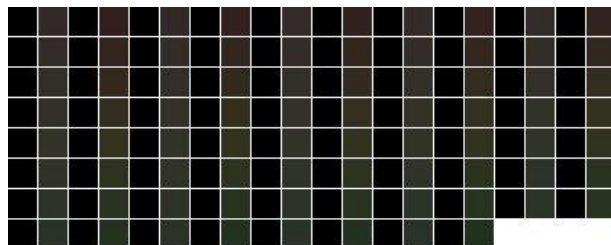
True Summer



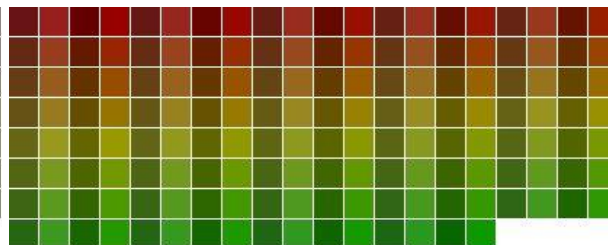
True Light



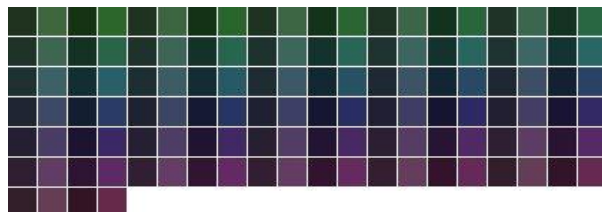
Deep Autumn



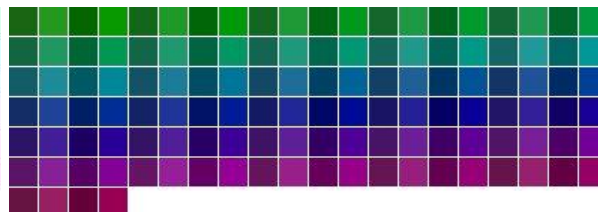
Soft Autumn



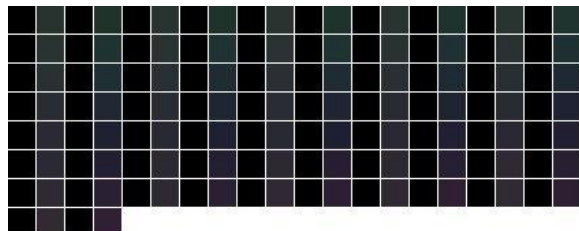
True Autumn



True Soft



Deep Winter



True Deep



True Winter

Bright Winter

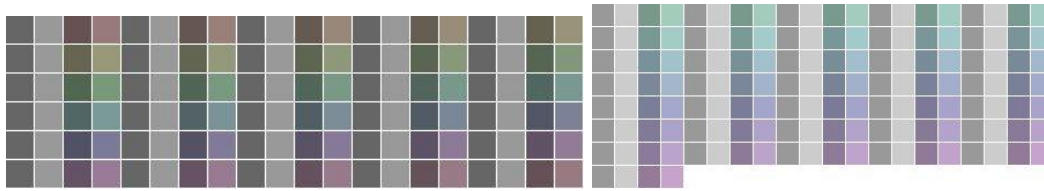


Fig. 5 - Fig. 20

The next step would be a recommendation of makeup products based on the color palettes from an existing database.

Product recommendation is done by comparing the generated seasonal palette's colors and the makeup products' respective colors from the database. For different makeup products, we have the following process:

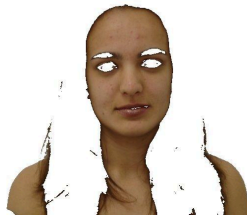
1. To recommend foundation, concealer, contour, and bronzer makeup products, we compare the average skin color with those products' existing shade in our database. If the exact shade is not found, we find the closest match.
2. To recommend lipstick, blush, eyeliner, and eyeshadow, we compare the average colors of lip and eye together with seasonal color palettes.
3. In addition, we also provide makeup product shades that contrast with the person's natural coloring.

5. Testing Results

Skin extraction



Original image 51-11F



Hue 0-30, SV 30-150



Hue 0-30, SV 40-80

SV range experiment



Original image 01-11



SV 45-105



SV 35-105

Hue and SV range combination experiment



Original image 01-11



Hue 3-21, SV 35-80



Hue 5-30, SV 29-170

Recursive Methods



Original image



Method 1

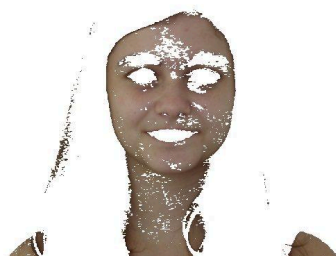


Method 2

Dark Lighting conditions



Original image



Hue 3-24, SV 30-96

Skin palette extraction



Method 1



Method 2



Method 3



Method 4



Eye and lip detection: Method 1

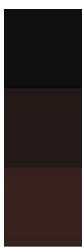
Original image



Extracted eyes, lips,
and eyebrows



Eye palette



Lip palette

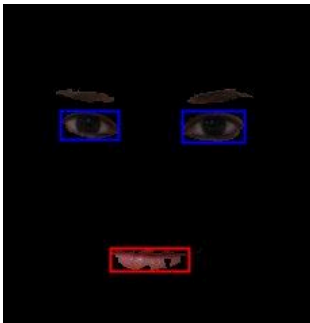


Eye and lip detection: Method 2

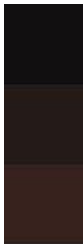
Original image



Detected eyes and lips



Eye palette



Lip palette



Hair extraction

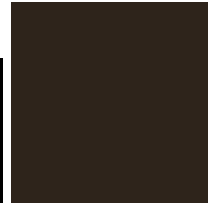
Original image



Extracted hair region



Average hair color



Final Results

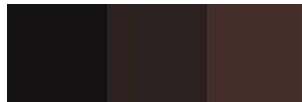
Original face image



Skin palette



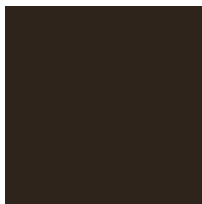
Eye palette



Lip palette

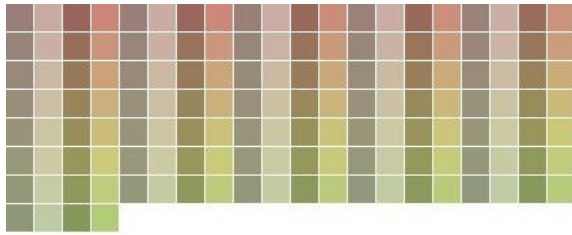


Hair color



Seasonal analysis: Warm Spring

Product recommended: Lipstick



Accuracy

To further improve the testing results, our goal will be to adapt our algorithms to perform seasonal analysis on another dataset (not only Brazilians).

We calculate the accuracy of creating a skin color palette from a face image to be the following:

1. Calculate the average hue of all extracted skin-like pixels.
2. Calculate the average hue of all hues derived from our generated skin color palette.
3. Find the closest match of those two averages and calculate the deviation of those averages.

6. Conclusion and Future Work

We plan to continue working on this research project to enhance its practical applications in real life, and improve its accuracy, in order to make it more user friendly. We are planning to emphasize a particular focus on developing a virtual try-on tool that will be the main key feature for making the project more user friendly and make people's lives easier. The real-time product visualization is a key direction for the future development of this project since it allows users to visualize how the recommended makeup products will appear on their face after completing the seasonal color analysis. The virtual try-on tool will simulate the application of the recommended makeup products on the user's facial image in real time or from a static photo. This will help the users to be more confident about the recommendation and their personal preferences before making purchases, which will result in saving resources, and make people more satisfied with their decisions. Thus, integrating virtual try-on technology would be a natural and highly beneficial extension of the current research. While the system already identifies a user's seasonal color type and recommends matching products, the ability to virtually try on the product before making a purchase will increase the user's trust. From a consumer perspective, the virtual try-on tool will contribute to greater confidence in product selection, minimize returns, and save time and money for both the consumers and producers.

From the technical perspective, the development of the virtual try-on feature will involve the use of augmented reality (AR), facial landmark detection, and image blending algorithms. For implementing these the main technologies and tools that will be explored, and if found suitable, used include:

1. OpenCV or Dlib for facial feature detection and landmark mapping. Both are libraries implemented in programming languages such as Python and C++ used for object detection and real-time computer vision.
2. For real-time facial tracking, MediaPipe or FaceMesh could be used. These are powerful tools developed by Google, which can help in facial features detection, and in placing the products on the right objects.
3. Since it would be comfortable for users to use the application on their smartphones, popular AR frameworks such as ARKit (iOS), ARCore (Android), or WebAR can be used for mobile/web deployment.
4. Additionally, blending techniques and alpha masking may be used to make the colors apply and overlay in a realistic way while preserving facial and product texture.

The initial methodology that should be researched to understand the effectiveness is that the tool should first use the facial feature extraction processes developed in this research to accurately segment skin, lips, eyes, and hair. Then the seasonal analysis should be given. Once the user receives personalized product recommendations based on the seasonal type given, the system will overlay the suggested shades directly onto their image or video stream. The interface will allow users to choose the exact recommended product and visualize the results instantly.

Another important step as our future work is developing a machine learning model using our results as a training dataset. The result that we have gathered can be a solid base for a model that will make the facial feature extraction more accurate for a larger scale of images. This will be a great contribution to the field of artificial intelligence and machine learning, especially for developing object detection, face recognition, and palette creation models. The model can then be further developed to conduct seasonal color classification and personalized recommendations.

To sum up, our research project introduces a systematic, data-driven seasonal color analysis based on a combination of color theory principles and image processing techniques. Through precise extraction of facial features such as skin, eye, lip, and hair colors, the system categorizes a person into one of the 16 seasonal color categories, eliminating the subjectivity and inconsistencies of conventional methods. The resulting personalized recommendations for makeup products demonstrate the practical value of this analysis, bridging scientific computation with real-world fashion and beauty applications.

Aside from its usefulness, the project is a significant contribution to computer vision, human-computer interaction, and personalized styling technology fields. Besides that, color analysis becomes more accessible to people. Instead of visiting a professional analyst, users can take a selfie, and our system would suggest their seasonal type and appropriate makeup shades.

The data produced by this project can be used as the basis for developing strong AI models and subsequently enhancing the ability of the system by using predictive modeling and supervised learning. The virtual try-on application will also allow users to see product suggestions in real time, which will improve the user experience and cosmetic confidence. In conclusion, this research not only drives technical methods in face extraction and color categorization but also sets the foundation for large-scale, user-centric applications in the beauty market, fashion tech, and beyond. It demonstrates how interdisciplinary innovation can empower individuals to make knowledgeable, confident choices regarding their appearance and how new possibilities are available for intelligent, inclusive, and interactive design. Ultimately, our project

brings art and technology together by taking the works of Itten, Caygill, and Munsell as a foundation while applying them in a modern, personalized way.

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